

OFUNATO, Japan

WMO: 475120

Lat: 39.0650N

Lon: 141.7150E

Elev: 38

StdP: 100.86

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.0	-3.9	-13.3	1.2	-1.9	-12.1	1.3	-1.3	9.7	1.5	8.7	1.1	2.3	340	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.4	30.6	23.7	28.8	23.1	27.4	22.5	24.6	28.5	24.0	27.5	23.4	26.5	3.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.6	18.5	26.2	23.0	17.8	25.9	22.4	17.1	25.5	74.9	28.8	72.6	27.6	70.2	26.5	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.1	7.0	6.2	DB	-7.0	33.8	1.4	1.6	-8.0	34.9	-8.9	35.9	-9.7	36.8	-10.7	38.0
			WB	-8.1	25.6	1.3	0.7	-9.1	26.1	-9.8	26.5	-10.6	26.9	-11.6	27.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	1.0	1.5	4.4	9.5	14.4	18.0	22.2	23.7	20.4	14.6	8.7	3.6
	DBStd	8.36	2.38	2.63	2.98	3.40	3.05	2.69	2.94	2.59	2.91	2.92	3.33	2.82
	HDD10.0	1015	279	239	176	50	3	0	0	0	0	2	64	200
	HDD18.3	2754	538	473	432	266	125	37	3	1	12	119	290	458
	CDD10.0	1702	0	0	2	34	141	241	378	425	311	145	24	1
	CDD18.3	400	0	0	0	1	4	28	123	167	73	4	0	0
	CDH23.3	2038	0	0	0	4	31	93	639	984	279	8	0	0
	CDH26.7	473	0	0	0	1	5	15	162	253	37	0	0	0
Wind	WSAvg	2.5	2.9	2.8	2.9	2.9	2.7	2.4	2.1	2.1	2.1	2.3	2.4	2.7
Precipitation	PrecAvg	1491	50	46	90	119	144	165	195	175	197	143	98	62
	PrecMax	1970	168	137	175	278	293	310	486	607	502	358	336	202
	PrecMin	1007	9	7	16	15	46	48	50	28	27	42	8	9
	PrecStd	224	43	32	45	65	64	64	104	132	102	71	76	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	12.4	17.3	22.9	27.0	28.8	32.8	33.3	29.7	24.2	19.2	14.3
		MCWB	6.2	7.4	9.8	12.7	15.9	21.0	24.0	24.7	23.2	18.8	13.8	10.8
	2%	DB	8.0	9.7	13.9	19.4	23.5	25.8	30.3	31.0	27.6	22.2	17.4	12.2
		MCWB	4.5	5.6	8.6	11.8	15.2	19.3	23.4	24.3	22.5	17.5	13.0	8.7
	5%	DB	6.5	7.7	11.7	17.3	21.4	23.9	28.4	29.2	26.1	20.8	16.0	10.2
		MCWB	3.3	3.9	7.2	10.7	14.4	18.4	22.9	23.8	21.9	16.8	12.2	7.1
	10%	DB	5.1	6.0	9.9	15.2	19.6	22.3	26.7	27.9	24.8	19.6	14.5	8.5
		MCWB	2.0	2.5	6.0	10.0	14.0	18.0	22.3	23.3	21.4	15.8	11.0	5.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	8.5	11.4	15.2	18.5	22.1	25.0	25.7	24.6	20.7	15.8	12.1
		MCDB	8.8	10.9	15.1	19.9	22.7	26.2	29.6	30.2	27.6	22.2	17.4	13.6
	2%	WB	5.2	6.2	9.5	13.2	17.0	20.6	24.2	24.9	23.7	19.1	14.2	9.4
		MCDB	7.4	9.1	12.6	16.9	20.2	24.0	28.2	29.1	26.1	21.2	16.5	11.4
	5%	WB	3.7	4.4	8.0	12.1	16.0	19.6	23.5	24.4	23.0	17.6	12.8	7.4
		MCDB	6.2	7.1	11.0	15.3	19.3	22.4	27.2	28.1	25.3	19.8	15.2	9.9
	10%	WB	2.4	2.8	6.5	11.1	15.0	18.7	22.9	23.9	22.0	16.4	11.5	5.7
		MCDB	4.9	5.5	9.6	14.1	18.4	21.4	26.0	27.1	24.3	18.8	13.9	8.2
Mean Daily Temperature Range	5% DB	MDBR	6.4	7.2	8.2	8.8	8.3	6.8	6.2	6.4	6.9	8.2	8.4	6.9
		MCDBR	8.3	9.7	11.1	12.0	11.4	9.4	8.9	8.4	7.7	8.6	10.2	9.5
		MCWBR	6.2	6.8	7.6	6.6	5.6	4.5	3.8	3.6	3.9	5.3	7.0	7.4
	5% WB	MCDBR	8.1	9.5	10.2	9.8	9.1	7.5	7.8	7.3	6.5	7.0	9.4	9.3
		MCWBR	6.6	7.3	8.1	6.3	5.4	4.0	3.7	3.4	3.8	5.3	7.3	7.7
Clear-Sky Solar Irradiance	taub	0.363	0.409	0.488	0.534	0.554	0.538	0.544	0.526	0.466	0.430	0.397	0.372	
	taud	2.171	2.037	1.886	1.836	1.851	1.971	2.028	2.076	2.208	2.195	2.187	2.182	
	Ebn at Noon	777	787	760	754	749	761	752	754	778	761	732	733	
	Edh at Noon	107	140	181	203	204	181	170	158	131	119	105	99	
All-Sky Solar Radiation	RadAvg	2.11	2.91	3.81	4.68	5.04	4.91	4.43	4.39	3.65	2.98	2.19	1.77	
	RadStd	0.13	0.17	0.21	0.42	0.48	0.32	0.60	0.56	0.33	0.29	0.14	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page